

The University of Azad Jammu and Kashmir, Muzaffarabad.



Computer Networks

Students Name:	Syed Abdulrehman Naseem
Roll No:	2024-SE-18
Course Code:	CS-2203
Lab Number:	04
Instructor:	Engr. Asma Javed

Department of Software Engineering

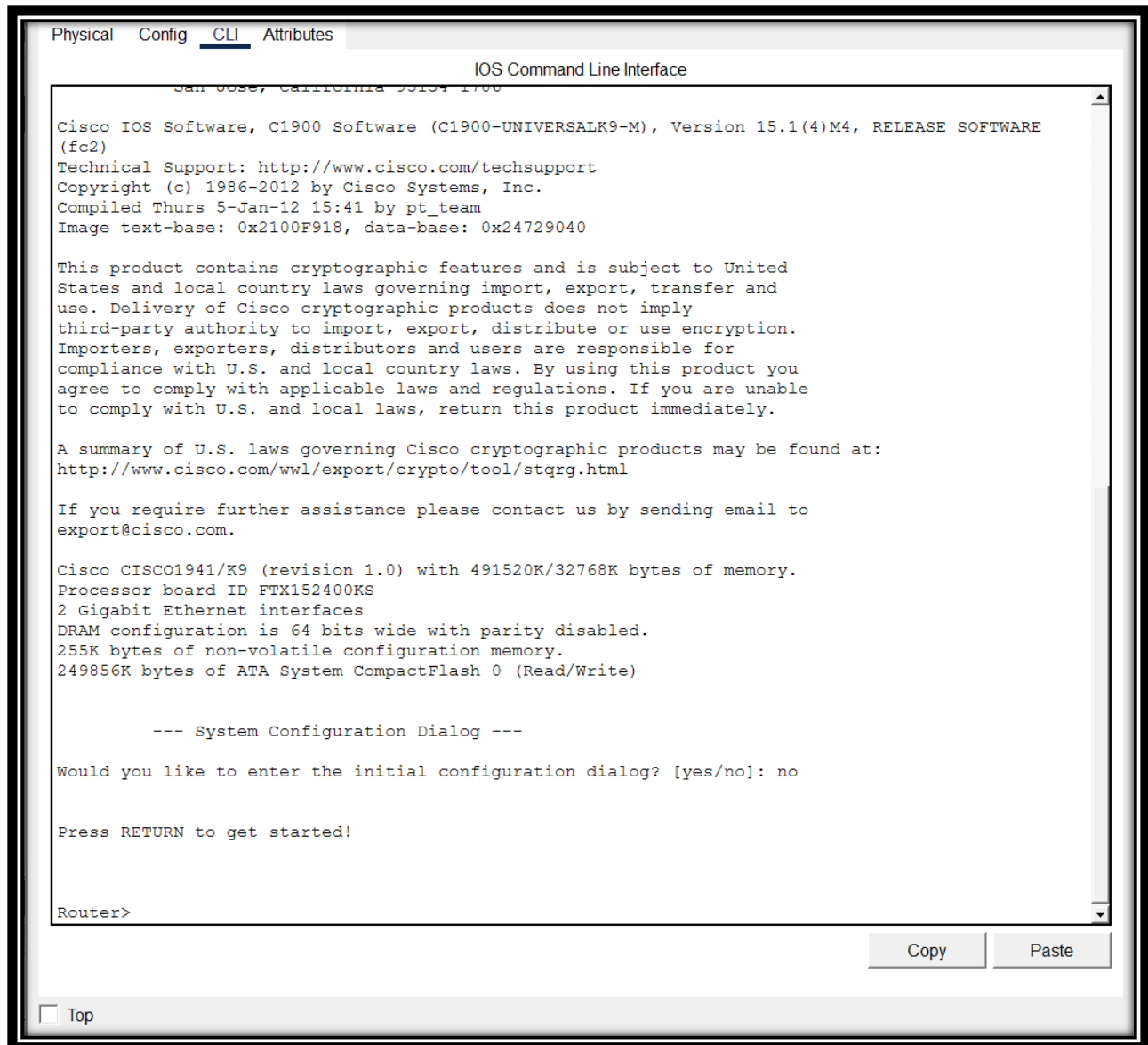
Router – CLI accession & configuration on Cisco Packet Tracer

Introduction

This lab introduces the basic method of accessing and configuring a Cisco router using the Command Line Interface. In computer networks, routers and switches must be configured before they can properly operate in a network. I used to simulate a real networking environment and practice router configuration commands. Screenshots are attached.

After opening Cisco Packet Tracer, I selected a router (1941).





This screenshot shows that the router CLI was opened successfully in Cisco Packet Tracer.

The **Router>** prompt indicates User EXEC Mode, which is the basic access mode.

```
--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>enable
Router#
```

Copy Paste

☐ Top

This screenshot shows that the enable command was used to move from User EXEC Mode to Privileged EXEC Mode. The Router# prompt confirms that privileged mode was entered successfully.

```
--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
```

Copy Paste

☐ Top

This screenshot shows that the router entered Global Configuration Mode. This mode is used to configure router settings such as passwords and security commands.

```
--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#enable password cisco123
Router(config)#service password-encryption
Router(config)#enable secret class123
Router(config)#
```

☐ Top

Copy Paste

This screenshot shows the basic security configuration of the router. An enable password was set, password encryption was enabled, and an enable secret password was configured for stronger security

```
--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no

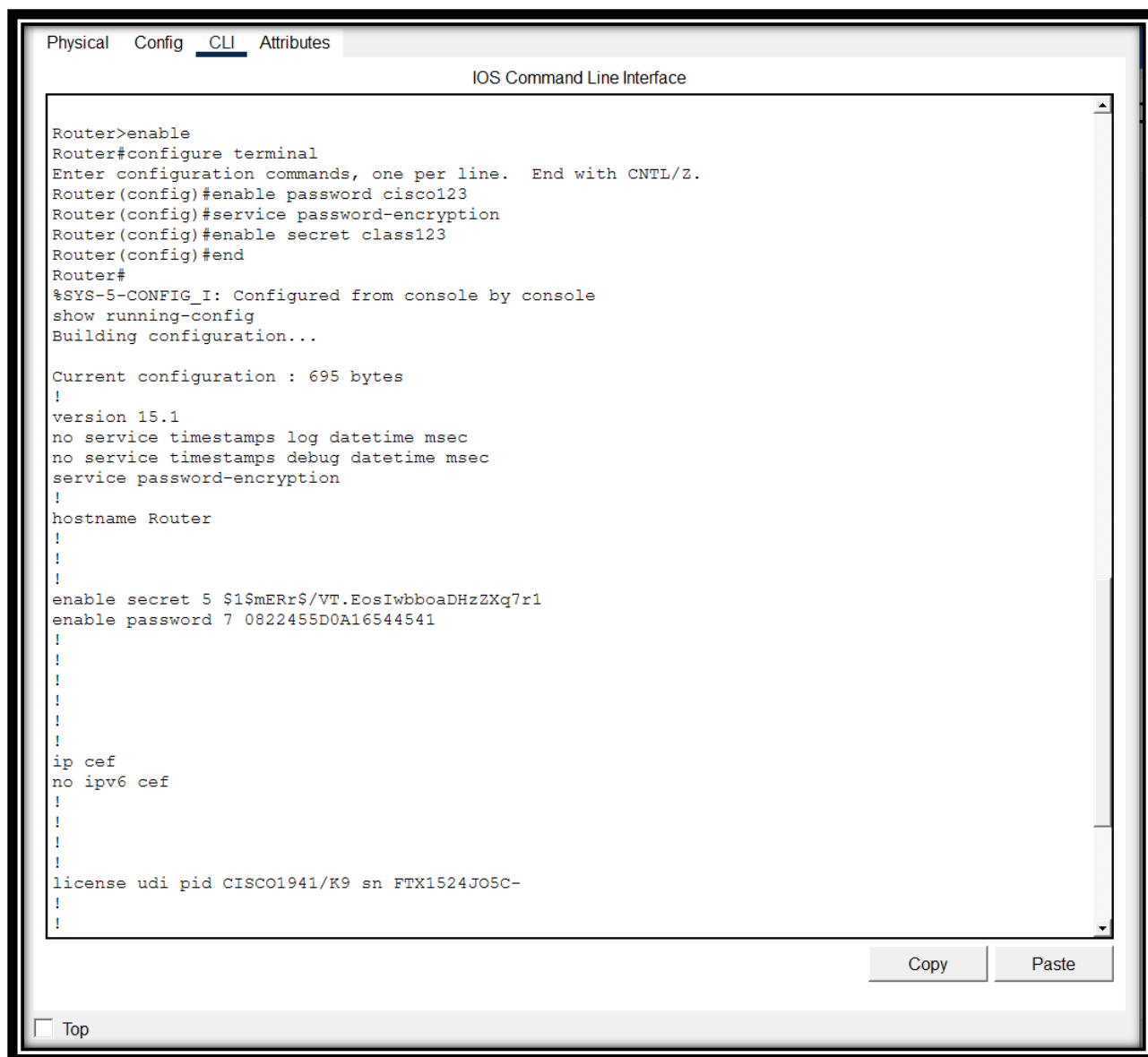
Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#enable password cisco123
Router(config)#service password-encryption
Router(config)#enable secret class123
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

☐ Top

Copy Paste

This screenshot shows that the end command was used to exit Global Configuration Mode and return to Privileged EXEC Mode.



The screenshot displays the 'CLI' tab of a network configuration tool. The main window shows the 'IOS Command Line Interface' with a terminal session. The user has entered 'enable' to enter configuration mode, followed by 'configure terminal'. They have then configured the password 'cisco123', enabled password encryption, and set a secret 'class123'. After exiting configuration mode, the user has entered 'show running-config' to display the current configuration. The output shows the configuration stored in RAM, including version 15.1, timestamps, password encryption, hostname 'Router', and the configured secret and password. At the bottom of the terminal window, there are 'Copy' and 'Paste' buttons. Below the terminal window, there is a 'Top' button with a checkbox.

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#enable password cisco123
Router(config)#service password-encryption
Router(config)#enable secret class123
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
show running-config
Building configuration...

Current configuration : 695 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname Router
!
!
!
enable secret 5 $1$mERr$/VT.EosIwbboadHZZXq7r1
enable password 7 0822455D0A16544541
!
!
!
!
!
ip cef
no ipv6 cef
!
!
!
!
license udi pid CISCO1941/K9 sn FTX1524JO5C-
!
!
```

☐ Top

This screenshot shows the running configuration of the router. The running configuration is stored in RAM and contains the active password and encryption settings.

```
!
line con 0
!
line aux 0
!
line vty 0 4
  login
!
!
!
end

Router#show startup-config
startup-config is not present
Router#
```

☐ Top

This screenshot shows the startup configuration before saving. As the startup-config is not present, it means the current router settings have not yet been saved to NVRAM.

```
Router#show startup-config
startup-config is not present
Router#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Router#
```

☐ Top

This screenshot shows that the running configuration was saved to the startup configuration. This ensures the router keeps the configured settings after restart.

```

Router#show startup-config
startup-config is not present
Router#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Router#show startup-config
Using 695 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname Router
!
!
!
enable secret 5 $1$mERr$/VT.EosIwbbodaDHzzXq7r1
enable password 7 0822455D0A16544541
!
!
!
!
!
!
ip cef
no ipv6 cef
!
--More-- |

```

This screenshot shows the startup configuration after saving. The saved configuration is stored in NVRAM and will be loaded when the router restarts.

Complete command list (In order)

- Enable
- configure terminal
- enable password cisco123
- service password-encryption
- enable secret class123
- end
- show running-config
- show startup-config
- copy running-config startup-config
- show startup-config

Conclusion

In this lab, I learned how to access a router CLI in Cisco Packet Tracer and navigate between User EXEC, Privileged EXEC, and Global Configuration modes. I configured basic router passwords, enabled password encryption, viewed running and startup configurations, and saved the configuration to NVRAM. Thank you!